

6. Temporary modulation for exercise and lighter (in-)activity V 4.0

Please note that with autoISF you are in an early-dev. environment, where the user interface is **not optimized for safety** of users who stray away from intended ways to use. Good safety features exist, but these are only as good as the development-oriented user understands and implements them.

This is not a medical product, refer to disclaimer in [section 0](#)



6.1 Dynamic iobTH and sensitivity ratio

6.1.1 Manual (direct) iobTH modulation

6.1.2 Automations for iobTH modulation

6.1.3 Dynamic iobTH

6.2 Temp. % profile switch

6.3 DIY cockpit based on User action Automations

6.4 Mastering the exercise after meal challenge

6.4.1 Manual mode

6.4.2 DIY cockpit button for User action Automation

6.4.3 Laissez-faire mode

6.5 Activity monitor based on step-counter

[Available related case studies:](#)

Case study 6.2: Biking day with hi carb lunch;
..... DIY cockpit

Preliminary remarks

For insights how to generally manage **various kinds of exercise**, listen-in this reference by looping pioneer and sportswoman Dana Lewis: https://bit.ly/DC1_631 (starts around minute 05:30), or read up in: <https://diyps.org/.../how-to-exercise-when-exercise-is...>

Note that **this** paper focusses on the challenges of exercise management in a **Full Closed Loop** system. It is **no easy read** because it attempts to describe *all options* to deal with *various types* of exercise.

Looking at case studies that relate to *your* kinds of exercise might be easier to digest than working your way through *all the options laid out* in this section.

- Please report *your* experience by publishing a case study.

33 Fortunately:

- 34 • If you do only light exercise, the Activity Monitor ([section 6.5](#)) might be all you need,
35 nothing else from this [section 6](#).
- 36 • For any of your (occasional or regular) exercise events, you can set *any* of them up at
37 your leisure, later, one at a time – and activate the one you need as you approach
38 exercise time.
- 39 • Also (and especially in your first weeks of FCL; with focus on meal management), you
40 can always **exit into Hybrid** Closed Loop or even Open Loop, and manage certain
41 exercise the way you always did before.

42

43 As long you were not able yet to define better ways, you should always be able to manage A
44 bg dropping during sports with **extra snacks** (keep those at hand).

45 Staying in contact with the related discord/github community should help greatly to find
46 suitable ways to manage *your* type(s) of exercise

47

48 6.1 Dynamic iobTH and sensitivity ratio in „exercise mode“

49

50 iobTH is a iob threshold you can set, **above which AAPS will no longer deliver** additional
51 **SMBs**.

- 52 • This overrides the SMB management via even/odd bg target differentiation).
- 53 • Regarding by how much “*the last SMB*” may shoot over iobTH, see [section 2.4](#).

54

55 **For exercise, we like to limit how high iob can go**, therefore automatic “dynamic”
56 reduction of your set iobTH (= iobMAX x iobTH%) is a benefit, notably as you can individually
57 tune it.

58 In autoISF 3.0 and later, a setting for iobTH is made in AAPS preferences, defined there as
59 fraction (e.g. 0.6) of your set maxIOB:

60 /OpenAPS_SMB/autoISF_settings/Full_Loop_settings: iob_threshold_percent,

61 => default iobTH = iobMAX x **iob_threshold_percent**

62

63 So, while iobTH could also be modulated via iobMAX, we mostly adapt the
64 iob_threshold_percent, to do that.

65

66 In the following, 3 principal avenues to temporary adjusting iobTH to your exercise
67 requirements are described:

- 68 • **Manual** intervention ([6.1.1](#))
- 69 • making use of **individually defined Automations** ([6.1.2](#)), and
- 70 • relying on the **automatic dynamic adjustments** coming with autoISF ([6.1.3](#)) .

71 The author experimented with all of them, but rarely needs manual intervention because the
72 *automatic “dynamic”* adjustments work pretty well, after some individual tuning (see e.g.
73 [case study 6.2](#)).

74 In any case, it is good to educate yourself about manual tweaking options, should the need
75 arise.

76

77 [6.1.1 Manual \(direct\) iobTH modulation](#)

78

79 „Manual“ routes to directly change iobTH would be

- 80 • changing the setting for the new parameter „iob_threshold_percent „
- 81 • or changing the setting for iobMAX

82 in /Preferences.

83 This is not a preferred route for temporary adjustment, because it is not easy accessible with
84 just a button stroke, and it would not automatically revert to your prior setting, after use. A
85 better solution that can achieve nearly the same is: to construct your own “DIY cockpit”
86 button to change iobTH% from the AAPS main screen, see next section.

87

88 [6.1.2 Automations for temporary iobTH modulation](#)

89

90 You can define Automations that set a different iobTH% **under pre-defined conditions**

91 In a variation of this idea (if your Automation has the User Action box ticked), you get a grey
92 button into your AAPS home screen, from which you can activate that changed iobTH
93 manually (“DIY cockpit”, as was already presented in [section 5.2.2.3](#))..

94 Note that your Automation defines the iobTH that your loop shall use **in place of** the
95 previously set iobTH:

- 96 ● it will still be modulated further if %profile and TT are set (see below)
- 97 ● it will overwrite the iobTH% you had set in /preferences!

98

99 **Caution:** A different iobTH% (same goes for bgAccel_ISF_weight) can *not* be set *temporarily* in
100 Automations (i.e. a *duration cannot be attached*). You **must** define a suitable **additional**
101 **Automation that** must be active *in tandem*, and **restores the prior set iobTH%** (or bgAccel-
102 ISF_weight) later **again**. Else, once your Automation sets in, it will *forever* shift this important
103 parameter setting!

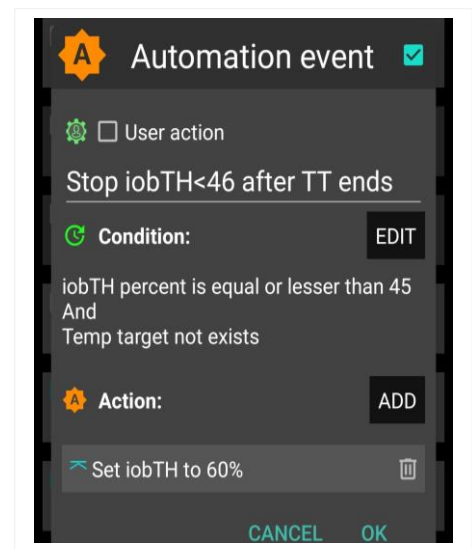
104 If for instance you have several Automations that, in combination with a set elevated TT also
105 set a lower iobTH: Don't be fooled, the duration only applies to the TT. You need an extra
106 “tandem” Automation for all of them.

107 Example: My Automation that restores my prior set profile
108 iobTH%:

109 I picked out the *highest* of the *lowered* iobTH values that
110 *any of my* Automations can set (45 percent was the
111 highest “of the low ones” in my case), and then I can
112 automatically restore to my *prior* 60% via this one:

112

113 **Caution:** Watch out for another potential stumbling block,
114 because many Automations only work under the condition
115 that no TT is already running.



115 As temp. changing iobTH is quite tricky to automate, it is the author's preferred route to only
116 *indirectly* modify it – see next section:

117

118

119

120

121 6.1.3 Dynamic iobTH: iobTH modulation via setting a temp. glucose target (TT)

122

123 In AAPS/Preferences, set “**High TT raises sensitivity** = TRUE”. Then, setting an **elevated**
124 temporary glucose target (TT), decreases iobTH by the same factor as it increases sensitivity
125 (as it “softens” ISF). Both measures decrease insulin that the loop will give.

126 Likewise, In AAPS/Preferences, set “**Low TT lowers sensitivity** = TRUE”. Then, setting a
127 **low** temporary glucose target (e.g. a EatingSoonTT of 74 mg/dl), elevates iobTH by the
128 same factor as it also sharpens (lowers) the ISF. The loop will give more insulin.

129

130 6.1.3.1 How does automatic sensitivity and iobTH adaptation work in the exercise mode?

131 **When. additionally. the exercise button is ON** (lit yellow), **iobTH gets reduced**
132 **particularly strong, and ISF is particularly weakened** (as desired for exercise).

133 That effect is the stronger (**ISF gets the weaker, iobTH the lower**), **the lower you set the**
134 **half-basal exercise target** for your exercise mode in AAPS/preferences/OpenAPS SMB:

135 The following table shows, for a profile target of 100 mg/dl, how the set ...

- 136 • half_basal_exercise_target you set in AAPS/preferences/OpenAPS SMB...

137 Choose a low number if you later want a high dynamic range of sensitivity modulation

138 Lower half-basal exercise target = lesser insulin delivered

- 139 • ...and current exercise TT (that you set on the day you do the respective exercise,
140 with an eye on how you wish sensitivity auto-adjusted)...

141 Higher TT = lesser insulin delivered

142 ... determine the effective sensitivity ratio:

Half basal ex.target	180	150	120
TT	sens.ratio	sens.ratio	sens.ratio
100 = profile target	1	1	1
120	0,8	0,71	0,5
140	0,67	0,56	0,33
160	0,57	0,45	0,25
180	0,50	0,38	0,20

143 The exact calculation for *any* combination of profile target, set TT, and half-
144 basal_exercise_target is given in the autoISF Quick guide (see [section 3.3](#)).

145 Note that:

146 • **temp. basal = profile basal * sens.ratio**

147 *Example: At a half-basal_exercise_target of 120, setting a TT of 120 gives only half (0.5) of profile*
148 *basal (hence the name of the parameter)*

149 • **temp.ISF = profile ISF / sens.ratio**

150 • **temp.iobTH = set iobTH * sens.ratio**

151 Whereas in “vanilla” AAPS the sens ratio is simply coming from you (manually), or from
152 Autosens (automatically) setting a temporary profile sensitivity other than 100% (and in the
153 special case of dynamicISF with additional effects on ISF), here, in autoISF, we have strong,
154 non-linear, and user scaleable effects on the sens.ratio.

155

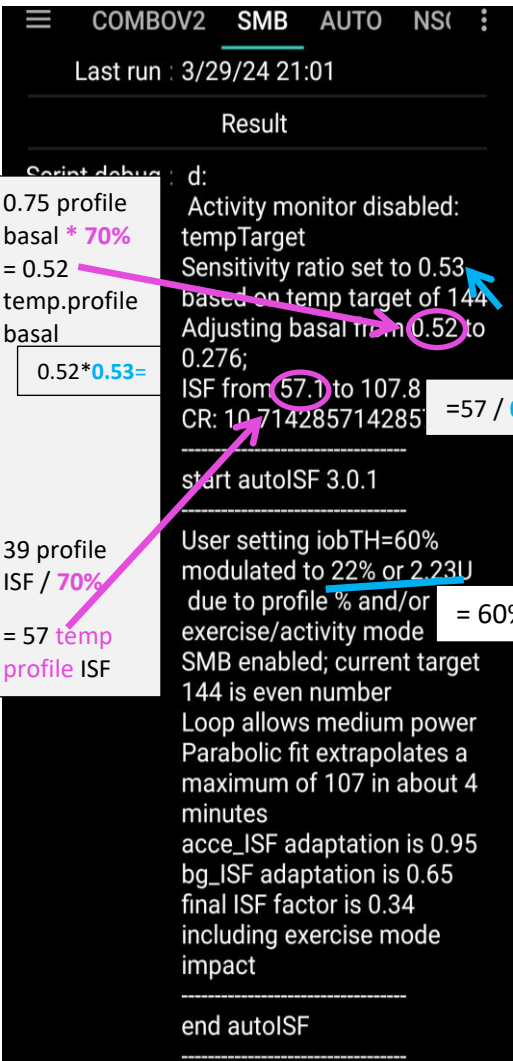
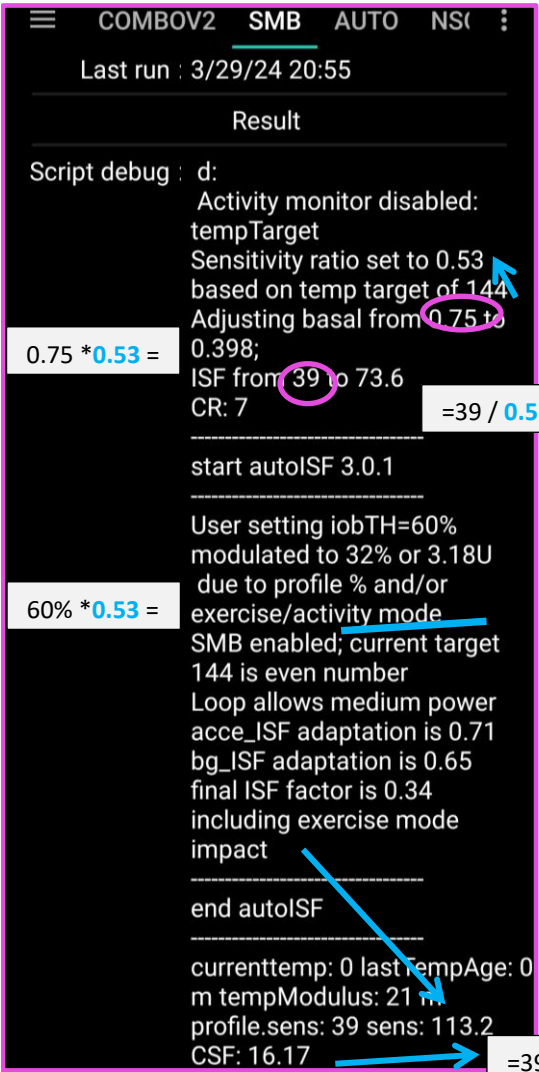
156 [6.1.3.2 How you recognize the real-time iobTH, and “aggressiveness” status of your FCL loop in](#)
157 [general](#)

158 Rather than bothering with the math, you can just look into your **SMB tab** where your
159 selected temporary settings put your iobTH, and the modified ISF (called **sens**):

160

The valid “effective iobTH” can also be seen in the SMB tab, see example (for a TT=144 and exercise button clicked);

Same, with (via top left button in AAPS home screen) **additionally 70% profile** applied:



profile.sens 39 sens: 162 = 39 / (0.34 * 70%)

These examples show that, **by just setting an exercise TT and a typical exercise profile%** (two super easy “interventions” via the top buttons on our AAPS main screen, turning yellow in response as an easy “reminder” we are in a special mode), **the iobTH will be automatically very sharply reduced**, in our example, to:

about half just by the TT; and further to about 1/3, by the % setting:

Exercise button ON + ...	... TT=144	... TT=144 AND 70% profile
iob TH = 6 U (60%)	iobTH = 3,18 U (32%)	iobTH =2,23 U (22%)
profileISF = 39 mg/dl/U	ISF = 73,6	ISF = 107
autoISF modulated ISF *)	ISF = 113,2	ISF = 162

*) data are from a situation where there is no sharp bg rise, so autoISF does not suggest increased aggressiveness

186 In the example fgiven above, the user's iobTH calculates to 6.0 U, which is 60% of iobMAX
187 of 10.0 U. So, normally, autoISF FCL could give SMBs until iob reached *anywhere between*
188 6.0 U *and* 7.2 U (=6.0 +20%; see [section 2.4](#) at step 2.4: at bg>100, iob can run max. 20%
189 over with "last" SMB).
190 For doing exercise, this window shrinks now to 3.18 – 3.82 U (left, TT-only) or, even to 2.23 –
191 2.77 U (right, TT AND %profile).

192 **In conclusion, these easy-to-make settings (TT, %profile) automatically provide the**
193 **same thing like would have been done in Hybrid Closed Loop, where a meal bolus of**
194 **about 7 U would get cut down to 4 U or even to 3 U, depending on type of exercise.**

195 If you concurrently use QPython 3L and the emulator on your Android phone (see [section 11](#))
196 you need not look into the SMB tab, but could see more details (~ for the last hour, plus all
197 contributing ...ISF_categories from autoISF), in tabular form, on your phone.

198 The most recent AAPS x autoISF version also allows extra graphs at the bottom of the AAPS
199 home screen on your smartphone, showing the development, and current value, of iobTH,

200 (Also details on bgAccel-, pp-, bg-, and dura-ISF contributions can be visualized in extra graph).

201 For i-Phone autoISF users, double clicking on "Statistics" provides some more detailed
202 information (see [section 11.3](#)).

203

204 6.1.3.3 Customization and-tuning

205 Try to determine good settings for the kinds of exercise that you engage in:

- 206 • Set your **half-basal exercise target** in /preferences that suits all of them...
- 207 • Determine reasonable TTs for each of your intended specific exercises
- 208 • Iterate through this a couple of times (whenever you happen to do *that*
209 exercise).

210 **Remember** ("code" for yourself), **which TT stands for which exercise type, so that just by**
211 **setting *that* TT everything (ISF, iobTH) *automatically* will provide the lower loop**
212 **aggressiveness that you need *for that specific type* of exercise.**

213

214

215

216

217 When setting a TT please watch out for unintended implications and side-effects:

218 (1) Setting a TT often **shuts out other** Automations.

219 Therefore, choose the **duration** wisely (and also the **sequence**, in which all your
220 Automations are listed).

221 (2) You always must consciously decide whether you set an **even** or an **odd**

222 numbered by **target** (TT or profile target). (This is assuming you use, as you
223 should, the even/odd by target differentiation for SMB on/off).

224 • 2.1) Pick **odd**, if you do not want SMBs during exercise. (Despite your softened ISF, and
225 lowered iobTH, SMBs still might „attack“ a sports snack too strongly).

226 ○ However, odd cannot be set too early, when your meal digestion still
227 requires SMBs.

228 ○ Likewise, you might want the option for a few automatically delivered SMBs
229 against unforeseen spikes (e.g. from excitement) also later.

230 In that case, an **Automation** might **sneak in a desired SMB** or two via
231 switching from odd to even, just for a couple of minutes, and under a well
232 thought-out set of conditions (that you must find in **your** data patterns,
233 when you do that kind of exercise that you try to find good settings for),

234 However, you are probably out of luck because an already set odd (or
235 any) TT would preclude such Automation from kicking in. Then you
236 need to **develop additional** ideas, another detour, like to first define an
237 **Automation that briefly shuts your odd TT down**.

238 ○ So, defining everything so you really can be happy with oddTT being your
239 primary way is a quite tricky project you should not under-estimate.

240 • 2.2) Working with an **even** TT can be preferable, notably of course if your exercise is one
241 that can get you totally excited, with glucose spikes.

242 ○ While this mode generally does allow SMBs, the loop softens the ISF (by the
243 sens.factor like in the table given above), and will temp. shut SMBs down, when
244 **iobTH** (which also got lowered by the sens. factor) is exceeded.

245

246 Whether odd or even TT is better depends on the kinds of exercise you are doing, and
247 probably depends on the protein and fat load of your meal and snacks, as well.

248 (3) **Timing** can be **critical** as to when you do your exercise announcement, especially
249 relative to a preceding hi-carb meal. Then you want the reduced iobTH in place latest after
250 you received the first SMB. See [section 6.4](#) and [case study 6.2](#)

251 (4) Once you are familiar with the **dynamic range of your iobTH**,

- 252 • after you made your settings, notably set your half-basal exercise target
- 253 • knowing the range of TTs and %profile adaptations that you intend to use
- 254 before/during/after your types of exercise

255 please confirm or re-consider *your iobTH_percent setting in /Preferences*, [section 2.4](#).

256 (5) You always can **look the effective iobTH up in the SMB tab** (see screenshots given 3
257 pages earlier). Also, it is given as a line in an extra graph that is available underneath your
258 main AAPS glucose screen.

259

260 6.1.4 Tweaking iobTH

261

262 You can use any of the above discussed methods, or also the one that now follows in [section](#)
263 [6.2](#), to *further tweak* iobTH temporarily, should you see a need.

264

265 Also outside of exercise, setting an **even elevated TT plus** pressing the **exercise button**
266 gives easy access **to significantly reduce aggressiveness of your autoISF loop** via a
267 resulting lowered iobTH and, concurrently, elevated effective ISF.

268 This could be used for instance for 45-60 minutes **at low/medium carb snacks**, as an
269 alternative to shutting SMBs **entirely** off via an **odd** TT.

270

271 When *exercise follows a meal*, it might be smart to use the just discussed tweaking methods
272 right after you felt the sting from the first big Lyumjev SMB.

273 However, we will look at smarter and safer ways for this “exercise after meal” scenario in
274 [section 6.5](#) and in [case study 6.2](#)

275

276

277 6.2 Temporary % profile switch

278

279 A complementary measure (to setting an elevated TT) that you also can take from the AAPS
280 home screen, is to set a **reduced temp.% profile** sensitivity.

281 This setting would **multiply** with the results in above table and further **reduce basal and**
282 **iobTH**_(whenever exercise button AND profile button both are yellow).

283 An example was already given with the 2nd screenshot, 4 pages earlier

284

285 Note that temp. reduction of basal will proportionally also **reduce the max. allowed size of**
286 **SMBs** (which is two hours worth of basal x SMB_range_extention, see [section 2.1](#))

287 This is desirable at exercise, albeit seldom needed, because milder ISFs will reduce the
288 requested SMB size, and after the radically lowered iobTH is exceeded, no SMBs will be
289 requested any longer, anyways.

290

291 The **time windows** for doing a profile switch *can differ* from the time window (duration) of
292 your TT-related exercise settings.

293 **Using all available tools then allows a nearly surgical approach to what you want to**
294 **achieve for and during your favorite exercise(s).**

295 • Often the %profile modulation is used for several hours if not days to accommodate “long
296 waved” sensitivity changes (See e.g. in [case study 6.2](#)).

297 • Instead, or even additionally, the percentage might be modified for just a couple of
298 minutes, or for one special snack or meal duration, to “nudge” the proportionally
299 modulated aggressiveness of the FCL (see [section 5.2.3](#)).

300

301 You can prepare yourself for anything you see coming up, or potentially coming up, in your
302 daily life, so, from the comfort of your cockpit ([section 6.3](#);) you get ready for it within just a
303 second or two, doing a few „clicks“.

304

305

306 6.3 DIY cockpit for exercise, based on User action Automations

307 6.3.1 Basic Settings for Exercise

308

309 Coming from FCL with no TT set (the top fields TT and exercise are grey), you best prepare
310 for an intended exercise by **pressing the TT field** of your AAPS main screen (your looping
311 cockpit; presented in [section 5.2](#)).

312 There, you can **freely select** TT and duration.

313

314 In case you want maximal “softening” effect on basal, ISF, and iobTH, activate also the
315 **exercise button** in the top middle of your AAPS main screen. It then should turn yellow, to
316 indicate you are in the dynamic exercise mode (color sequence of the top 3 buttons in your
317 AAPS home screen = YY $\underline{Y}$ or GY $\underline{Y}$).

318

319 6.3.2 „Dynamic“ exercise mode off = traditional AAPS exercise mode (YGY)

320

321 When the dynamic exercise mode is off (YGY), you still have the instruments for *exercise*
322 *management just as you always had it in the past* = a combination of manually softened
323 aggressiveness via setting a temp. %profile change, and orienting corrections towards an
324 elevated TT.

325 By selecting an odd numbered TT you now have the *additional option* to shut SMBs
326 temporarily off, too.

327 In this “YGY” mode, the % temp. set profile is the applied “effective sensitivity” (% ratio)

328 TT and duration can be entered or changed (= traditional mode to set exercise targets).

329 The effective iobTH is given in the SMB tab, or can also be seen in an extra graph below the
330 AAPS home screen glucose chart.

331 TT and % profile will also show on the yellow labels of the neighboring %profile (left top of
332 AAPS home screen) and TT (right side), respectively.

333 The middle (exercise) field remains grey because the automatic sensitivity tuning (that would
334 use TT and half-basal exercise target) are off.

335

337

341

348

354

359

360

361 6.4 Mastering Exercise after a Meal

362

363 In Hybrid Closed Loop, we gave less insulin at meals (a reduced bolus) before exercise.

364 Since we now get our meal insulin automatically from the loop, we would have to at least
365 somehow tell it that exercise follows this time.

366 Simply setting an exercise profile *before* the meal would make our full closed loop too weak
367 in the "treatment" of the first glucose rise. **What we want is, to get our** (already, compared
368 to HCL, delayed) **meal insulin delivered as fast as possible by SMBs. It just should be**
369 **capped at the desired iob reduction.**

370

371 6.4.1 Manual mode requires 2 user interventions

372

373 What we can do, is (1) **reduce** the **iobTH** (via the `_%` setting, *e.g. by one third*).

374 • *In the example we were using, this would mean to reduce by 2 U to $iobTH^* = 4U$.*

375 • Do that estimate for your data, and think back how you did bolus reduction in hybrid
376 closed loop before same exercise.

377 • Likewise, you can use your profile ISF, *e.g. 30 mg/dl/U* and „translate“ by how much
378 ($2U * 30 \text{ mg/dl/U} = 60 \text{ mg/dl}$) this „pulls you away from going into a hypo“.

379 • Using your IC (*e.g. 8g/U*) you can also translate the iobTH reduction (2 U) into a
380 „snack equivalent“ ($2U * 8 \text{ g/U} = 16 \text{ g}$) that you „replace“ by thinking ahead and
381 „budgeting“ for some exercise with your iobTH modulation.

382 In this senario, our loop delivers SMB insulin as fast as always, only that when the last SMB
383 has passed the iobTH, the loop only has elevated %TBR to work with, meaning it cannot
384 raise iob by much any longer. This provides an elevated glucose level on which we enter
385 exercise, and saves us hypo danger or snack need (as calculated in above examples).

386

387 After this reduced iobTH is reached (or up to 30% exceeded by the last SMB, up to 20% @
388 even $TT > 100 \text{ mg/dl}$), step (2) must follow = an increased exercise **bg target** is set (see
389 [section 6.2](#)).

390

391 The problem with this approach is that it requires **two** user interventions, first **setting the**
392 **lower iobTH%**, and later (**and this *in a time-critical manner***, after iobTH is exceeded), to
393 **input an exercise TT**, or to activate a related setting.

394 To eliminate this problem, the following refined solutions are suggested:

395

396 6.4.2 DIY cockpit: Using pre-set meal / exercise settings from a User action Automation

397

398 The „DIY cockpit“ user interface allows a *one-step* setting for meal + exercise that can be
399 selected in time-uncritical fashion, any time before the meal starts (or even shortly thereafter,
400 within the “grace period” after which we hope to already see the first SMB triggered).

401 A detailed example is given in [case study 6.2](#):

402 A sequence of 3 Automations must be set up, of which only the first one must be manually
403 triggered, in just one time-uncritical key stroke from the AAPS home screen.

404 The others are activated automatically, when the respective Conditions are met.

405

406 Automation #1 provides, for a meal that precedes exercise, the full loop aggressiveness, but
407 makes sure that this aggressiveness stops immediately after a (reduced) iobTH is exceeded.
408 The reduced iobTH ensures that not too much insulin is on board for exercise after the meal.
409 Also it provides an elevated bg level at (re-)start of exercise.

410

411 In this Automation, the box “User action” should be permanently ticked. This will
412 automatically provide a **grey button on the bottom of the AAPS home screen** (“DIY
413 cockpit”) that can be freely named (= headline of Automation #1).

414 For exercise that is not done frequently, I choose to get rid of that cockpit button by disabling
415 the Automation fully, in my list of Automations... until the evening before e.g. a bike tour, when
416 I will want to have my cockpit give me the optional button again.(See [case study 6.2](#))

417

418 As soon as the (reduced) iobTH is exceeded, two things need to be provided :

419 (1) a milder running FCL (reduced exercise %profile, after the meal rise had been
420 managed based on 100% profile boosted further by bgAccel_ISF driven full loop
421 aggressiveness) => Automation #2 sets e.g. 70% profile and ends TT

422 (2) setting an exercise TT (not possible with Automation #2. But *after* it terminated the
423 TT, an Automation #3 can immediately follow, and set the desired exercise TT=125
424 (which implies the exercise mode

425 Note that Automations 2 and 3 are fully automatic, no User action is involved. See [case](#)
426 [study 6.2](#) for an example

427

428 Should, during the exercise, a need arise to modulate the loop aggressiveness (jobTH,
429 effective ISF), this can be done within 1-2 seconds, also right from the AAPS home screen
430 („FCL cockpit“), by setting a higher or lower temp. %profile, and/or by setting a higher or
431 lower temp. exerciseTT.

432 To make the loop temporarily act a bit more aggressive, switching the exercise button OFF
433 (from yellow to grey) could also be considered

434

435 **Defining User action - Automations to build your FCL cockpit**

436

437 If you want to develop *your* **DIY User Interface**, make sure you define suitable settings that
438 reflect ***your*** personal insulin sensitivity and data patterns.

439

440 **Caution:** As mentioned in other places, Automations can be tricky as to whether they
441 actually will ever work, because the loop goes through the exact **sequence of all your**
442 **active Automations**, and might be switched into a direction that no longer is compatible with
443 the conditions that must be a given, for the Automation you think that should kick in.

444

445 To have a clean AAPS home screen (and also to prevent unnecessary/accidental activation
446 by kids playing around with offered buttons), define reasonable time windows for each of
447 your shelved special routines, or keep them entirely dormant (de-activated) in the list of all
448 your Automations, and activate them only for/on the day when you think you might need
449 them

450

451

452

453 6.4.3 Laissez-faire alternative

454

455 You could make your life easier: **Just use** (as in Hybrid Closed Loop) **an exercise setting**
456 and accept a resulting reduced loop aggressiveness **already before meal start**. You would
457 go a bit higher in your glucose peak. As, in principle, a higher glucose level is desirable for
458 starting exercise, this can be a viable route, too, **especially if you do a** (often
459 recommended) ***protein-rich meal before exercise***.

460 Logic: For a high carb containing meal, we wanted in the preceding sections a strong initial
461 FCL response, but only up to a certain (lowered) iob. (This resembles the reduced user bolus
462 in HCL around exercise.).

463 However, the more our meal shifts to high protein/low carb, and the more we are accepting of
464 bg going a bit higher before we start or resume our exercise, the better we can tolerate a
465 reduced FCL aggressiveness also at meal start, i.e. the entire day.

466

467

468

6.5 Activity Monitor

An optional feature for times without serious exercise, but still suspected **effects on insulin sensitivity** is the **activity monitor**.

It can be generally activated under /preferences/OpenAPS SMB/Activity modifies sensitivity)

If the user

- has scaling factors set there (in preferences/OpenAPS SMB/Activity modifies sensitivity)

- has **no TT running**

- (and, regarding nighttime: did not opt for „ignore_inactivity_overnight“)

then AAPS automatically modulates for sensitivity changes **based on movement intensity**

for the last minutes to 1 hour time frame.

Loopers whose insulin sensitivity is affected by erratic patterns of total in-activity (desk, couch) and moving around (walking, light house and garden work) can greatly benefit from this feature. Responding in a ~ 60 minutes (rather than in 8 – 24 hours) time frame makes the Activity Monitor far superior to Autosens.

Personalized tuning of the **two scaling factors** is necessary *in your FCL set-up phase*.

For details see ~p.9 of the “autoISF 3.##. Quick guide” in <https://github.com/ga-zelle/autoISF>

Later, *in your running FCL*, this **will automatically adjust insulin delivery** (basal, ISF, and iobTH; see 1st screen of AAPS SMB tab!) to suit activity state of the past minutes (up to 1 hour). See also [section 5.1.5](#).

Effects from the Activity Monitor are hard-limited to go **maximally**

- **plus 20%** insulin at detected resistance (in-sensitivity to insulin) **at in-activity**
- **minus 30%** insulin at detected increased sensitivity to insulin due to **activity**.

495 **Note that Activity Monitor only works when no exercise (or other) TT is active:**

- 496 • Whenever you set a TT, you consciously go for a certain effect on the sensitivity ratio
497 to be used *in that time window*.

498 Usually it will be stronger than the tweaking that the Activity Monitor would do.
499 So, you would not want the Activity Monitor interfere, and additionally tweak
500 things you just defined differently for a certain situation, and time window, by
501 setting a TT.

- 502 • During the set TT, your Activity Monitor **keeps collecting** the data on your activity/in-
503 activity. Immediately after the set TT ends, the Activity Monitor **automatically**
504 **resumes** its work

505 This is good news also for those who might use brief even/odd target switches
506 (e.g. when sneaking-in a small snack w/o triggering a SMB), but would hate to
507 see their Activity Monitor function go under for a while afterwards.

508

509 You can easy, in real-time, check the impact of your Activity Monitor on the sensitivity used
510 (to calculate your insulinRequired, or also to auto-adjust also iobTH) in your AAPS **SMB tab**.
511 From autoISF 3.0.1 onwards, this is super easy to retrieve in the 1st screen, on top of the
512 autoISF results.

513